



Homework 9: This homework discusses the scattering amplitude $f(\theta)$ for a two-body system. Starting from the two-body Hamiltonian and separating it into center-of-mass and relative motion, we will see how this leads to outgoing spherical waves at large distances. The bonus questions explore a scattering problem with particles that are identical, and how the symmetrization or anti-symmetrization of the wavefunction enforces selection rules.

Take two particles of equal mass $m_1 = m_2 = m$ interacting through a central potential

$$V = V(|\mathbf{r}_1 - \mathbf{r}_2|), \quad (1)$$

The full Hamiltonian is

$$\hat{H} = \frac{\hat{\mathbf{p}}_1^2}{2m} + \frac{\hat{\mathbf{p}}_2^2}{2m} + V(|\mathbf{r}_1 - \mathbf{r}_2|). \quad (2)$$

We want to show that, in the center-of-mass frame, the wavefunction of relative motion takes the standard scattering form

$$\psi(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k} \cdot \mathbf{r}} + f(\theta) \frac{e^{ikr}}{r}. \quad (3)$$

a) Define center-of-mass and relative coordinates and their conjugate momenta:

$$\mathbf{R} \equiv \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2), \quad \mathbf{r} \equiv \mathbf{r}_1 - \mathbf{r}_2, \quad \mathbf{P} \equiv \mathbf{p}_1 + \mathbf{p}_2, \quad \mathbf{p} \equiv \frac{1}{2}(\mathbf{p}_1 - \mathbf{p}_2). \quad (4)$$

Verify that $[\mathbf{R}_i, \mathbf{P}_j] = i\delta_{ij}$ and $[\mathbf{r}_i, \mathbf{p}_j] = i\delta_{ij}$, while the mixed commutators $[\mathbf{R}_i, \mathbf{p}_j]$ and $[\mathbf{r}_i, \mathbf{P}_j]$ vanish. This shows that $(\mathbf{R}, \mathbf{P})$ and $(\mathbf{r}, \mathbf{p})$ are two independent canonical pairs.

b) Rewrite the kinetic energy of Eq. (2) in terms of $\mathbf{P}$ and $\mathbf{p}$, and show that the Hamiltonian looks like

$$\hat{H} = \frac{\hat{\mathbf{P}}^2}{2M} + \frac{\hat{\mathbf{p}}^2}{2\mu} + V(r), \quad \text{with} \quad M = 2m, \quad \mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{m}{2}, \quad r = |\mathbf{r}|. \quad (5)$$

Here, the total mass is M and the reduced mass is μ . The central potential depends only on the relative coordinate by assumption.



This means that since $(\mathbf{R}, \mathbf{P})$ and $(\mathbf{r}, \mathbf{p})$ are canonical pairs, $\mathbf{P} = -i\nabla_{\mathbf{R}}$ and $\mathbf{p} = -i\nabla_{\mathbf{r}}$, so $\mathbf{P}^2 = -\nabla_{\mathbf{R}}^2$ and $\mathbf{p}^2 = -\nabla_{\mathbf{r}}^2$.

c) Because $\mathbf{P}$ commutes with $\hat{H}$, we can look for simultaneous eigenstates of $\hat{H}$ and $\mathbf{P}$. Substitute the wavefunction

$$\Psi(\mathbf{R}, \mathbf{r}) = e^{i\mathbf{K} \cdot \mathbf{R}} \psi(\mathbf{r}), \quad (6)$$

into Schrödinger's equation $\hat{H}|\Psi\rangle = E_{\text{tot}}|\Psi\rangle$ and show that $\psi(\mathbf{r})$ satisfies

$$\left[-\frac{\nabla_{\mathbf{r}}^2}{2\mu} + V(r) \right] \psi(\mathbf{r}) = E \psi(\mathbf{r}), \quad E \equiv E_{\text{tot}} - \frac{\mathbf{K}^2}{2M}. \quad (7)$$

Now we will be concerned with the center-of-mass frame, $\mathbf{K} = \mathbf{0}$, in which E is the total kinetic energy of relative motion and the lab and CM scattering angles coincide.

d) For a scattering state with energy $E = k^2/(2\mu)$, and for well outside the range of V , the function $\psi(\mathbf{r})$ satisfies the free Helmholtz equation

$$(\nabla^2 + k^2)\psi(\mathbf{r}) = 0. \quad (8)$$

Show that an incoming plane wave $e^{i\mathbf{k} \cdot \mathbf{r}}$ is a solution, and that the spherical wave e^{ikr}/r is also a solution. Explain in words why both pieces are needed in the asymptotic form Eq. (3). Why

does the scattering amplitude depends only on the zenith angle θ (the angle between $\hat{\mathbf{r}}$ and $\hat{\mathbf{k}}$) $f = f(\theta)$? Give at least one reason why $f(\theta)$ cannot be completely spherically symmetric, i.e., why f cannot be a constant independent of θ .

e) Compute the probability current

$$\mathbf{j} = \frac{1}{2\mu i} (\psi^* \nabla \psi - \psi \nabla \psi^*) \quad (9)$$

for the outgoing spherical wave $\psi_s(\mathbf{r}) = f(\theta) e^{ikr}/r$ alone, and show that the radial component at large r is

$$j_r^s = \frac{k}{\mu} \frac{|f(\theta)|^2}{r^2} + \mathcal{O}(r^{-3}). \quad (10)$$

Compare with the incident flux $j_i = k/\mu$ from the plane wave and give an expression for the differential cross section $d\sigma/d\Omega$ in terms of $f(\theta)$.



2) Bonus question on identical particles.

When the two scattering particles are identical, the detector at angle θ cannot tell which particle is which. Exchanging the two particles in the CM frame corresponds to $\mathbf{r} \rightarrow -\mathbf{r}$, which in spherical coordinates means $(\theta, \phi) \rightarrow (\pi - \theta, \phi + \pi)$. The total wavefunction must therefore be either symmetric (bosons) or antisymmetric (fermions) under this exchange.

a) Show that under $\mathbf{r} \rightarrow -\mathbf{r}$ the asymptotic form Eq. (3) becomes

$$\psi(-\mathbf{r}) \sim e^{-i\mathbf{k}\cdot\mathbf{r}} + f(\pi - \theta) \frac{e^{ikr}}{r}, \quad (11)$$

and construct the (anti)symmetrized spatial wavefunction

$$\psi_{\pm}(\mathbf{r}) = \frac{1}{\sqrt{2}} [\psi(\mathbf{r}) \pm \psi(-\mathbf{r})]. \quad (12)$$

Identify the symmetrized scattering amplitude $f_{\pm}(\theta) \equiv f(\theta) \pm f(\pi - \theta)$.

b) Because the two particles are indistinguishable, a detector at angle θ and a detector at angle $\pi - \theta$ count the same physical event. Taking proper account of this to avoid double-counting, the differential cross section is

$$\left. \frac{d\sigma}{d\Omega} \right|_{\pm} = |f(\theta) \pm f(\pi - \theta)|^2. \quad (13)$$

Expand this expression to show that it differs from the distinguishable-particle result by an *interference term*:

$$\left. \frac{d\sigma}{d\Omega} \right|_{\pm} = |f(\theta)|^2 + |f(\pi - \theta)|^2 \pm 2 \operatorname{Re}[f^*(\theta) f(\pi - \theta)]. \quad (14)$$

Show that at $\theta = \pi/2$ the symmetric (bosonic) cross section is twice the classical (distinguishable) result, while the antisymmetric (fermionic) cross section *vanishes*.

c) The scattering amplitude can be expanded in partial waves as

$$f(\theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} (2\ell + 1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos \theta), \quad (15)$$

where δ_{ℓ} are the phase shifts and P_{ℓ} are the Legendre polynomials. Using $P_{\ell}(\cos(\pi - \theta)) = P_{\ell}(-\cos \theta) = (-1)^{\ell} P_{\ell}(\cos \theta)$, show that

$$f(\theta) \pm f(\pi - \theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} [1 \pm (-1)^{\ell}] (2\ell + 1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos \theta). \quad (16)$$

What selection rule does this imply for the allowed partial waves ℓ in the scattering of identical bosons? How about identical fermions?

Solutions



Homework 9.

a) Using $[\mathbf{r}_{ai}, \mathbf{p}_{bj}] = i\delta_{ab}\delta_{ij}$ for $a, b \in \{1, 2\}$:

$$[\mathbf{R}_i, \mathbf{P}_j] = [\frac{1}{2}(\mathbf{r}_{1i} + \mathbf{r}_{2i}), \mathbf{p}_{1j} + \mathbf{p}_{2j}] = \frac{1}{2}(i\delta_{ij} + i\delta_{ij}) = i\delta_{ij}, \quad (17)$$

$$[\mathbf{r}_i, \mathbf{p}_j] = [\mathbf{r}_{1i} - \mathbf{r}_{2i}, \frac{1}{2}(\mathbf{p}_{1j} - \mathbf{p}_{2j})] = \frac{1}{2}(i\delta_{ij} - i\delta_{ij}) = i\delta_{ij}. \quad (18)$$

The mixed commutators vanish, e.g.

$$[\mathbf{R}_i, \mathbf{p}_j] = [\frac{1}{2}(\mathbf{r}_{1i} + \mathbf{r}_{2i}), \frac{1}{2}(\mathbf{p}_{1j} - \mathbf{p}_{2j})] = \frac{1}{4}(i\delta_{ij} - i\delta_{ij}) = 0, \quad (19)$$

and similarly $[\mathbf{r}_i, \mathbf{P}_j] = 0$. So $(\mathbf{R}, \mathbf{P})$ and $(\mathbf{r}, \mathbf{p})$ are two independent canonical pairs.

b) From $\mathbf{p}_1 = \frac{1}{2}\mathbf{P} + \mathbf{p}$ and $\mathbf{p}_2 = \frac{1}{2}\mathbf{P} - \mathbf{p}$,

$$\frac{\mathbf{p}_1^2 + \mathbf{p}_2^2}{2m} = \frac{\mathbf{P}^2}{4m} + \frac{\mathbf{p}^2}{m} = \frac{\mathbf{P}^2}{2M} + \frac{\mathbf{p}^2}{2\mu}, \quad (20)$$

with $M = 2m$ and $\mu = m/2$. Since $V(|\mathbf{r}_1 - \mathbf{r}_2|) = V(r)$ depends only on the relative coordinate, the full Hamiltonian takes the separated form Eq. (5).

c) Because H splits into a piece depending only on $\mathbf{P}$ and a piece depending only on $(\mathbf{r}, \mathbf{p})$, with no cross terms, the Schrödinger equation decouples into two independent equations for the CM and relative wavefunctions. Note that the CM equation is a free-particle equation with plane wave solution $e^{i\mathbf{K}\cdot\mathbf{R}}$ and energy $\mathbf{K}^2/(2M)$. Substituting $\Psi(\mathbf{R}, \mathbf{r}) = e^{i\mathbf{K}\cdot\mathbf{R}}\psi(\mathbf{r})$:

$$\frac{\mathbf{K}^2}{2M} e^{i\mathbf{K}\cdot\mathbf{R}}\psi(\mathbf{r}) + e^{i\mathbf{K}\cdot\mathbf{R}} \left[-\frac{\nabla_{\mathbf{r}}^2}{2\mu} + V(r) \right] \psi(\mathbf{r}) = E_{\text{tot}} e^{i\mathbf{K}\cdot\mathbf{R}}\psi(\mathbf{r}). \quad (21)$$

Cancelling $e^{i\mathbf{K}\cdot\mathbf{R}}$ and defining $E \equiv E_{\text{tot}} - \mathbf{K}^2/(2M)$ gives Eq. (7), a single-particle Schrödinger equation for a fictitious particle of mass μ moving in $V(r)$. In the CM frame ($\mathbf{K} = \mathbf{0}$), E is the entire kinetic energy of relative motion, and lab and CM scattering angles coincide because the CM is at rest.

d) For r much greater than the range of V , Eq. (7) reduces to

$$(\nabla^2 + k^2)\psi = 0, \quad k^2 \equiv 2\mu E. \quad (22)$$

A plane wave $e^{i\mathbf{k}\cdot\mathbf{r}}$ with $|\mathbf{k}| = k$ trivially satisfies this, since $\nabla^2 e^{i\mathbf{k}\cdot\mathbf{r}} = -k^2 e^{i\mathbf{k}\cdot\mathbf{r}}$. For the spherical wave, the hint gives

$$\nabla^2 \left[\frac{e^{ikr}}{r} \right] = \frac{1}{r} \frac{d^2}{dr^2} (e^{ikr}) = \frac{-k^2 e^{ikr}}{r}, \quad (23)$$

so e^{ikr}/r is also an exact solution of the Helmholtz equation. The source of the spherical wave is at $r = 0$. Multiplying by an angular factor $f(\theta)$ introduces extra terms from the angular Laplacian that fall off as $1/r^3$ and are subleading at large r , so $f(\theta)e^{ikr}/r$ also satisfies the Helmholtz equation in the asymptotic region to the order that matters.

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Why must V fall faster than $1/r$? The whole logic above relies on dropping $V(r)$ in Eq. (7). For an outgoing spherical wave e^{ikr}/r , what really matters is the phase accumulated by the outgoing wave between r and infinity due to the potential. With this insight, we can use a trick to show when the asymptotic ansatz Eq. (3) is justified (see, e.g., D. Tong's notes). We can recall a result from WKB theory that the phase acquired by the wavefunction as the potential varies slowly is just an integral of the local momentum $k(r) = \sqrt{2\mu(E - V(r))}$. Calculating this for radii larger than a given radial reference value r , we get

$$\phi(r) \simeq \int_r^\infty dr' (k(r') - k) \simeq -\frac{\mu}{k} \int_r^\infty dr' V(r'), \quad (24)$$

where we used $k(r') = \sqrt{2\mu(E - V(r'))} \simeq k - \mu V(r')/k$. For the asymptotic ansatz $\psi \simeq e^{i\mathbf{k}\cdot\mathbf{r}} + f(\theta)e^{ikr}/r$ to make sense, we need $\phi(r) \rightarrow 0$ as $r \rightarrow \infty$, i.e. the integral must converge.

If $V(r) \sim 1/r^n$ at large r , then $\int_r^\infty dr'/r'^n$ converges only for $n > 1$. This is precisely the statement that V must fall off *faster* than $1/r$. For $V \sim 1/r$ exactly (the Coulomb case), the integral diverges logarithmically and the outgoing wave acquires a phase $\sim (\mu/k) Ze^2 \ln(2kr) = \eta \ln(2kr)$, where η is called the Sommerfeld parameter. The asymptotic Coulomb wavefunction is therefore not e^{ikr}/r but $e^{i[kr - \eta \ln(2kr)]}/r$, and the entire scattering formalism has to be reformulated to accommodate this logarithmic distortion (distorted waves). For potentials falling faster than $1/r$ (Yukawa, finite range, $1/r^2$, etc.), the standard ansatz Eq. (3) is justified and $f(\theta)$ is well-defined.

e) The solution form Eq. (3) is motivated by the fact that the Helmholtz equation has two linearly independent solutions, a plane wave and a spherical wave:

$$\psi(\mathbf{r}) \xrightarrow{r \rightarrow \infty} e^{i\mathbf{k}\cdot\mathbf{r}} + f(\theta, \phi) \frac{e^{ikr}}{r}. \quad (25)$$

The first term shows the wave is prepared with definite momentum $\mathbf{k}$ at $r \rightarrow -\infty$ along the beam axis. The second represents the scattered wave with scattering amplitude f . It encodes all the information about how V has “redistributed” flux into the various directions.

For a central potential $V = V(r)$, the only vector available to construct f is $\hat{\mathbf{k}}$. That is, only the angle θ between $\hat{\mathbf{r}}$ and $\hat{\mathbf{k}}$ is meaningful.

f) For the incoming plane wave $\psi_{\text{inc}} = e^{i\mathbf{k}\cdot\mathbf{r}}$,

$$\mathbf{j}_{\text{inc}} = \frac{1}{2\mu i} (e^{-i\mathbf{k}\cdot\mathbf{r}} i\mathbf{k} e^{i\mathbf{k}\cdot\mathbf{r}} - \dots) = \frac{\mathbf{k}}{\mu}, \quad (26)$$

with magnitude $|\mathbf{j}_{\text{inc}}| = k/\mu$, the relative speed.

For the scattered piece $\psi_{\text{sc}} = f(\theta) e^{ikr}/r$, the radial derivative is

$$\partial_r \psi_{\text{sc}} = f(\theta) \left[\frac{ik e^{ikr}}{r} - \frac{e^{ikr}}{r^2} \right] = ik \frac{f(\theta) e^{ikr}}{r} + \mathcal{O}(r^{-2}). \quad (27)$$

Substituting into $j_r = (2\mu i)^{-1} (\psi_{\text{sc}}^* \partial_r \psi_{\text{sc}} - \dots)$,

$$j_r^{\text{sc}} = \frac{1}{2\mu i} \left[\frac{f^*(\theta) e^{-ikr}}{r} \cdot ik \frac{f(\theta) e^{ikr}}{r} - \dots \right] + \mathcal{O}(r^{-3}) = \frac{k}{\mu} \frac{|f(\theta)|^2}{r^2} + \mathcal{O}(r^{-3}), \quad (28)$$

which is Eq. (10). The angular components of ∇ acting on ψ_{sc} produce contributions smaller by an explicit $1/r$ and can be neglected.

The differential cross section is the ratio of the scattered flux through the area element $r^2 d\Omega$ to the incident flux:

$$\frac{d\sigma}{d\Omega} = \frac{r^2 j_r^{\text{sc}}}{|\mathbf{j}_{\text{inc}}|} = |f(\theta)|^2, \quad (29)$$

The cross terms between $e^{i\mathbf{k}\cdot\mathbf{r}}$ and $f(\theta)e^{ikr}/r$ do not enter $d\sigma/d\Omega$ as long as $\theta > 0$.

(Bonus question)

a) Under $\mathbf{r} \rightarrow -\mathbf{r}$, we have $|\mathbf{r}| = r$ and the angle between $\mathbf{k}$ and $-\mathbf{r}$ is $\pi - \theta$. Therefore

$$\psi(-\mathbf{r}) \sim e^{-i\mathbf{k}\cdot\mathbf{r}} + f(\pi - \theta) \frac{e^{ikr}}{r}. \quad (30)$$

Constructing $\psi_{\pm} = (\psi(\mathbf{r}) \pm \psi(-\mathbf{r}))/\sqrt{2}$:

$$\psi_{\pm}(\mathbf{r}) \sim \frac{1}{\sqrt{2}}(e^{i\mathbf{k}\cdot\mathbf{r}} \pm e^{-i\mathbf{k}\cdot\mathbf{r}}) + \frac{1}{\sqrt{2}}[f(\theta) \pm f(\pi - \theta)] \frac{e^{ikr}}{r}, \quad (31)$$

so the symmetrized scattering amplitude is $f_{\pm}(\theta) = f(\theta) \pm f(\pi - \theta)$.

b) We played with the normalization of our waves and the scattering amplitude, so let's check the normalization of the cross section. We will need the incident and scattered flux. Putting $\mathbf{k} = k\hat{\mathbf{z}}$ for simplicity,

$$j_i = \frac{1}{2\mu i} (\psi_{\pm}^* \nabla \psi_{\pm} - \psi_{\pm} \nabla \psi_{\pm}^*) = \frac{1}{2\mu i} \left[\frac{(e^{-ikz} \pm e^{ikz})ik(e^{ikz} \pm e^{-ikz})}{2} - \dots \right] = \frac{k}{\mu}, \quad (32)$$

$$j_r^s = \frac{1}{2\mu i} (\psi_{\pm}^* \partial_r \psi_{\pm} - \dots) \simeq \frac{1}{2\mu i} \left[\frac{|f_{\pm}(\theta)|^2 ik}{r^2} - \dots \right] = \frac{k}{\mu} \frac{|f_{\pm}(\theta)|^2}{r^2}. \quad (33)$$

where we dropped terms of order r^{-3} from the angular derivatives. Overall, we have

$$\left. \frac{d\sigma}{d\Omega} \right|_{\pm} = \frac{r^2 j_r^s}{j_i} = \frac{r^2}{j_i} \frac{k}{\mu} \frac{|f_{\pm}(\theta)|^2}{r^2} = |f_{\pm}(\theta)|^2. \quad (34)$$

No extra factor of $1/2$ is needed thanks to a cancellation between the normalization of incident and scattered waves. Using that

$$\left. \frac{d\sigma}{d\Omega} \right|_{\pm} = |f_{\pm}(\theta)|^2 = |f(\theta)|^2 + |f(\pi - \theta)|^2 \pm 2 \operatorname{Re}[f^*(\theta)f(\pi - \theta)]. \quad (35)$$

The first two terms are the distinguishable contribution and sum over the two possibilities “particle 1 detected at θ ” and “particle 2 detected at θ ,” the latter meaning particle 1 emerged at $\pi - \theta$. The third term is a genuinely quantum interference between direct and exchange amplitudes.

At $\theta = \pi/2$, $f(\theta) = f(\pi - \theta) \equiv f(\pi/2)$, so

$$\left. \frac{d\sigma}{d\Omega} \right|_{+, \pi/2} = 4|f(\pi/2)|^2, \quad \left. \frac{d\sigma}{d\Omega} \right|_{-, \pi/2} = 0. \quad (36)$$

The distinguishable baseline at $\pi/2$ is $|f|^2 + |f|^2 = 2|f(\pi/2)|^2$, so identical bosons give twice this, while identical fermions give exactly zero. =

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Before moving on, let me also mention that the total cross section $\sigma = \int d\Omega d\sigma/d\Omega$ in the identical particle case has a subtlety. In the way we defined $d\sigma/d\Omega$ in Eq. (13), we are counting the same physical event twice, once at θ and once at $\pi - \theta$. To avoid this double-counting, we can restrict the integration to $\theta \in [0, \pi/2]$ and multiply by 2 to account for the other half of the solid angle:

$$\sigma_{\pm} = 2 \int_0^{2\pi} d\phi \int_0^{\pi/2} d\theta \sin \theta |f(\theta) \pm f(\pi - \theta)|^2. \quad (37)$$

This is equivalent to integrating over the full solid angle but including a factor of $1/2$ in the definition of $d\sigma/d\Omega$ to avoid double-counting, which is the more common convention in the literature. This factor of $1/2$ is called the *symmetry factor* and is often denoted S . For two identical particles, $S = 1/2$; for distinguishable particles, $S = 1$; for three identical particles, $S = 1/3!$, etc.

c) Inserting Eq. (15) into $f(\theta) \pm f(\pi - \theta)$ and using $P_{\ell}(-\cos \theta) = (-1)^{\ell} P_{\ell}(\cos \theta)$:

$$f(\theta) \pm f(\pi - \theta) = \frac{1}{k} \sum_{\ell=0}^{\infty} [1 \pm (-1)^{\ell}] (2\ell + 1) e^{i\delta_{\ell}} \sin \delta_{\ell} P_{\ell}(\cos \theta). \quad (38)$$

The bracket $[1 + (-1)^{\ell}]$ vanishes for odd ℓ and equals 2 for even ℓ . Therefore:

- Identical spinless bosons: only $\ell = 0, 2, 4, \dots$ contribute.
- Identical fermions: only $\ell = 1, 3, 5, \dots$ contribute.

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Note! These statements refer only to the spatial part of the wavefunction. If you have a pair of fermions in the spin singlet state, the spatial wavefunction is symmetric and only even ℓ contribute, just like for bosons. In addition, note that spin-1 bosons can be in the spin triplet, which is antisymmetric (recall, $|1, 0\rangle = \frac{1}{\sqrt{2}}(|1\rangle_1 | -1\rangle_2 - | -1\rangle_1 | 1\rangle_2)$), so they can have odd ℓ as well.